

**Web-Based Resource Borrowing and Returning System
with Availability Tracking and Faculty Authentication
for MIS Faculty at ICCT Cainta Campus**

Complete Documentation — Based on Deployed System

<https://icct-mis.classapparelp.com>

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Abstract

The borrowing and returning of resources at ICCT Cainta Campus currently relies on manual request letters or faculty ID presentation, which is inefficient, time-consuming, and lacks transparency in tracking availability. Professors often experience delays in borrowing, while MIS faculty struggle with monitoring overdue returns and managing resource quantities. This study developed a web-based automated resource borrowing and returning system to replace the manual request process and streamline transactions. Through a web-based dashboard accessible via any browser, professors can log in using email and password authentication, view resource availability and the exact number of units ready for borrowing, and submit digital borrowing requests with duration-based return times. Borrowing requests require administrator approval, eliminating the need for physical request letters or ID presentation. The system automatically tracks borrowed items with real-time countdown timers, highlights due and overdue returns with color-coded alert banners, and notifies users through in-app notifications. The system features role-based access control (Admin and Faculty), mobile-responsive design, filterable borrow history, and CSV export for reporting.

I. INTRODUCTION

A. Background and Motivation

Resource borrowing and returning is an essential support function in academic institutions, particularly for faculty members who rely on shared materials and equipment to deliver instruction effectively. At ICCT Cainta Campus, professors in the MIS department currently request resources either by submitting a formal request letter or by presenting their faculty ID. While this process ensures accountability, it is slow, manual, and prone to delays. MIS faculty also face challenges in monitoring overdue returns and tracking the availability of resources, as there is no centralized system that provides real-time visibility of inventory. These inefficiencies disrupt schedules, reduce productivity, and complicate resource management.

B. Problem Statement

The existing borrowing and returning process at ICCT Cainta Campus is inefficient and outdated. Professors must either submit a request letter or present their faculty ID to borrow resources, which consumes time and lacks transparency. MIS faculty struggle to monitor overdue returns and resource availability, as there is no centralized dashboard to track inventory in real time. These limitations result in delays, poor accountability, and reduced efficiency in resource management.

C. Objectives of the Study

The general objective of this study is to design and develop a web-based system that automates the borrowing and returning of resources. Specifically:

1. Provide a web-based dashboard with email/password authentication and role-based access (Admin and Faculty).
2. Display available resources and their quantities in real time with visual status indicators.
3. Allow professors to submit digital borrowing requests by selecting a duration (1 hour, 2 hours, 3 hours, or custom).
4. Enable administrators to approve/reject requests and monitor due/overdue returns with real-time alert banners and countdown timers.
5. Provide filterable borrow history with status-based tabs (All, Pending, Approved, Borrowed, Returned, Rejected).
6. Generate downloadable CSV reports of all borrowing transactions.
7. Send automated in-app notifications with unread count badge.

D. Scope of the Study

- **Target Users:** MIS faculty and administrators at ICCT Cainta Campus.
- **Web-Based Access:** Accessible via browser URL on any device. Fully responsive (phone card layout, tablet compact table, laptop full table).
- **Authentication:** Email/password login with role-based access control and password recovery.
- **Real-Time Resource Management:** Live display of available resources and quantities.
- **Duration-Based Borrowing:** Digital requests with preset (1h, 2h, 3h) or custom (HH:MM) duration. Auto-calculated due date.
- **Administrative Oversight:** Approve/reject requests, mark returned, monitor due/overdue items with color-coded countdown.
- **Notifications:** In-app notifications with red badge on bell icon. Types: approved, rejected, due soon, overdue.
- **Borrow History:** Filterable by status with summary statistics.
- **Reporting:** CSV export for all borrow records.
- **Overdue Alert:** Red banner with specific item details and due dates.
- **Technical Platform:** Laravel 11, MySQL, Apache 2.4, Tailwind CSS, Alpine.js, Vite, Let's Encrypt SSL.

E. Limitation of the Study

- Requires internet connectivity.
- Password recovery depends on registered email access.
- Requires modern web browser on any device.
- Excludes physical maintenance of resources.
- Deployed for MIS faculty at ICCT Cainta Campus only.
- In-app notifications accessible only when logged in.
- Email notifications limited to account verification.

II. METHODOLOGY

A. Research Design

This study adopted a developmental research design focusing on systematic analysis, design, and implementation of an information system. The methodology emphasized iterative development with continuous stakeholder feedback.

B. Technical Stack

Component	Technology
Backend Framework	Laravel 11 (PHP)
Database	MySQL (icct_mis_db)
Frontend	Tailwind CSS + Alpine.js
Build Tool	Vite
Web Server	Apache 2.4 (Let's Encrypt SSL)
Email Service	Brevo (SendinBlue) SMTP
Server OS	Ubuntu 22.04 LTS on DigitalOcean
Domain	icct-mis.classapparelp.com

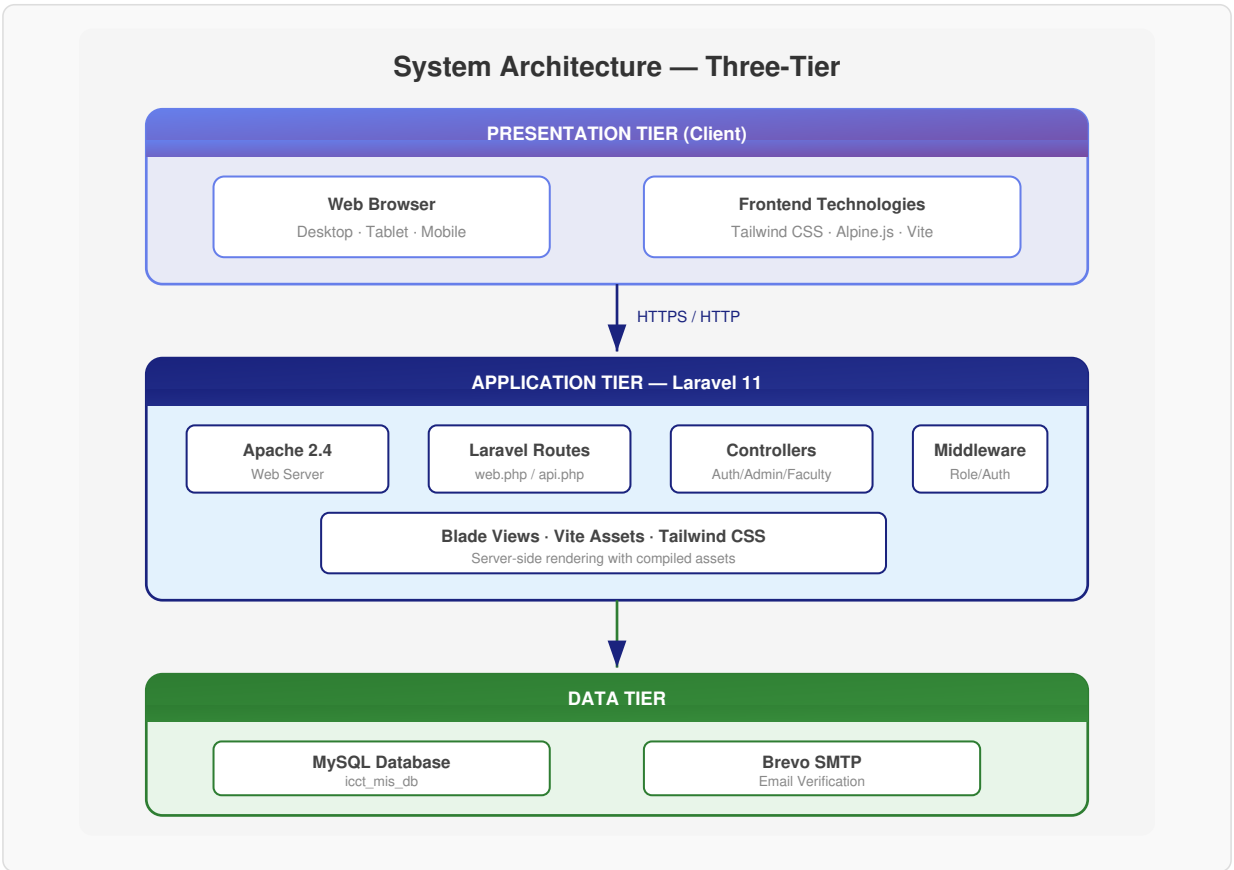
C. System Architecture

The system follows a three-tier architecture:

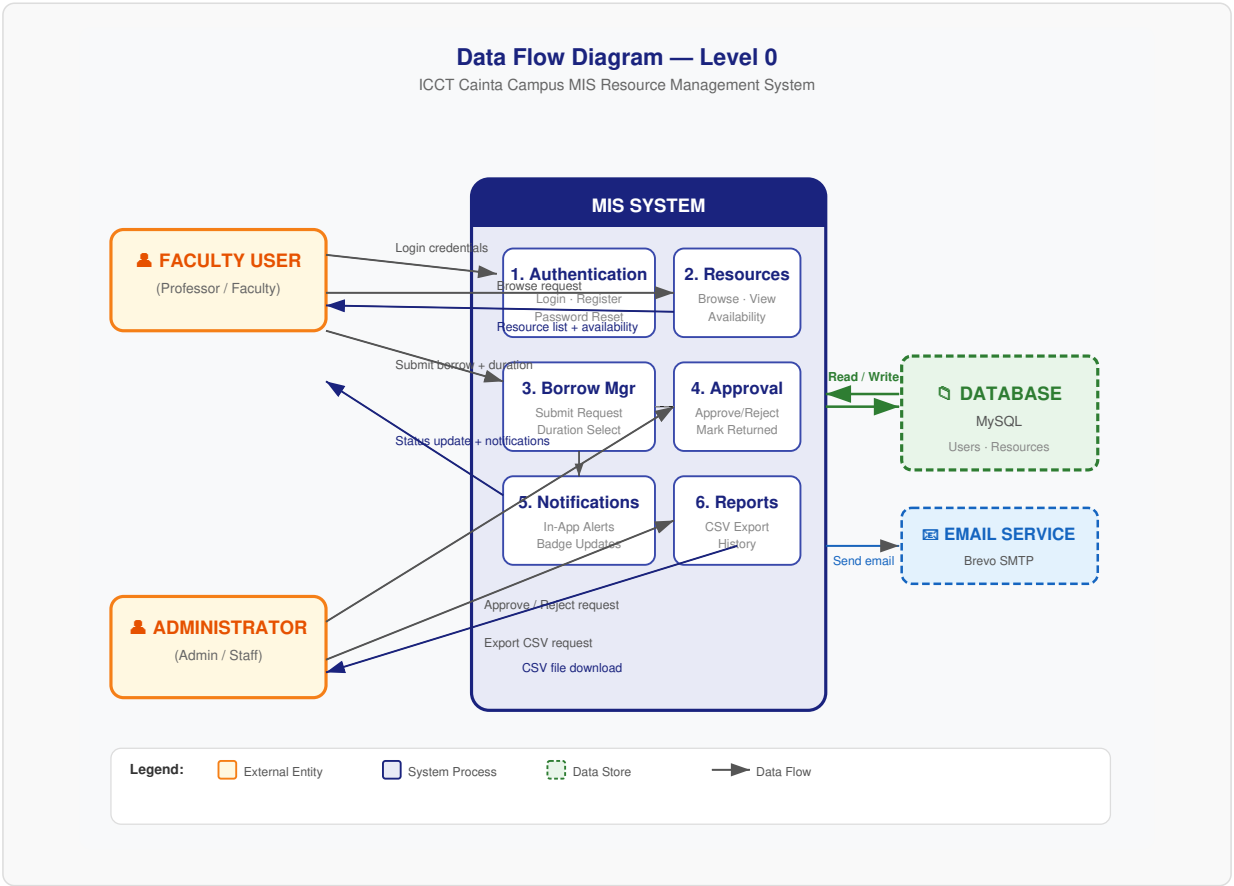
- 1. Presentation Tier:** Web browsers render Blade templates with Tailwind CSS and Alpine.js. The interface automatically adapts to desktop, tablet, and mobile screens.
- 2. Application Tier:** Laravel 11 handles routing, business logic via controllers, authentication via middleware, and server-side rendering of Blade views with Vite-compiled assets.
- 3. Data Tier:** MySQL stores all transactional data including users, resources, borrows, and notifications. All timestamps use UTC with Asia/Manila timezone conversion for display.

III. SYSTEM DIAGRAMS

1. System Architecture Diagram

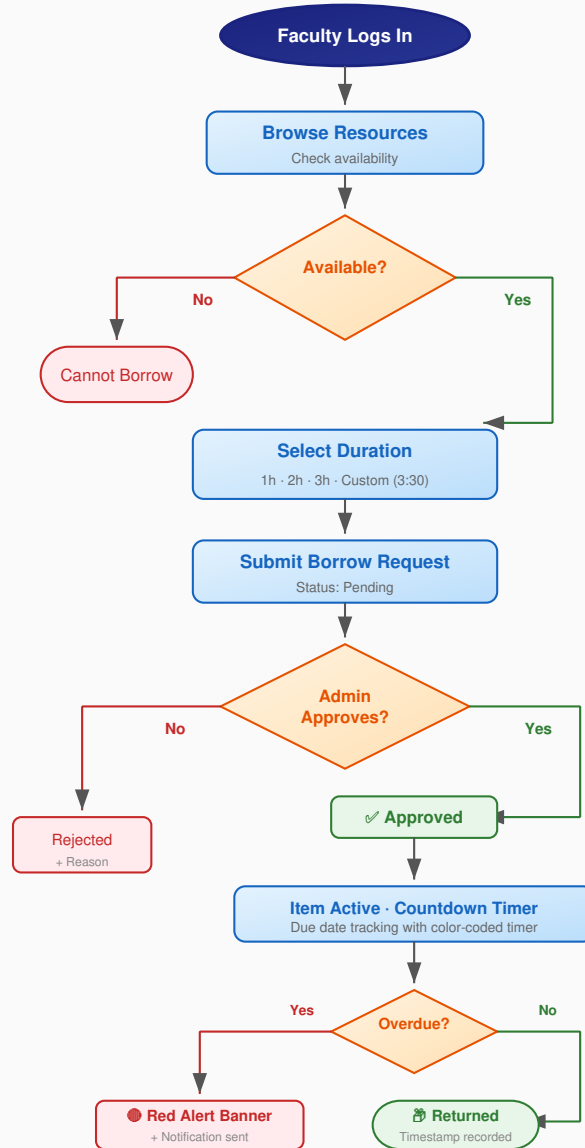


2. Data Flow Diagram (DFD Level 0)



3. Borrow Process Flowchart

Borrow Process Flowchart



IV. SYSTEM FEATURES

1. **Authentication:** Login/Register with role selection, password show/hide, password reset, email verification via Brevo.
2. **Role-Based Access:** Admin middleware, faculty scoped data, separate dashboards.
3. **Resource Management:** CRUD with auto-calculated availability, desktop table + mobile cards.
4. **Borrowing Workflow:** Duration select (1h/2h/3h/custom), auto due-date, approve/reject, mark returned.
5. **Dashboard:** Faculty overview + stats + overdue banner. Admin analytics + charts.
6. **Notifications:** In-app with red badge, delete/clear all, mark all read.
7. **Overdue Monitoring:** Cron job every minute, countdown timers, red alert banner.
8. **Borrow History:** Filter tabs, summary stats, mobile layout.
9. **CSV Export:** Admin download of all records.
10. **Mobile Responsive:** 3-tier layout (phone cards, tablet compact, laptop full).

V. CONCLUSION

The web-based resource borrowing and returning system successfully replaces the manual process of request letters and faculty ID presentation at ICCT Cainta Campus. The system provides real-time resource availability tracking, digital borrowing requests with duration-based return times, administrative approval workflows, automated overdue detection with alert banners, and comprehensive borrow history with filter capabilities. The system is deployed and accessible at <https://icct-mis.classapparelph.com>.

VI. RECOMMENDATIONS

1. QR code scanning for quick resource page access
2. Email notifications for borrow alerts
3. Advanced analytics with charts
4. Resource categorization and label printing
5. SMS notifications
6. Multi-department scalability
7. Mobile application version